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VIGIL

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International Data Science Olympiad

An AI-Driven Platform for Deforestation Risk Prediction in Vietnam

Background

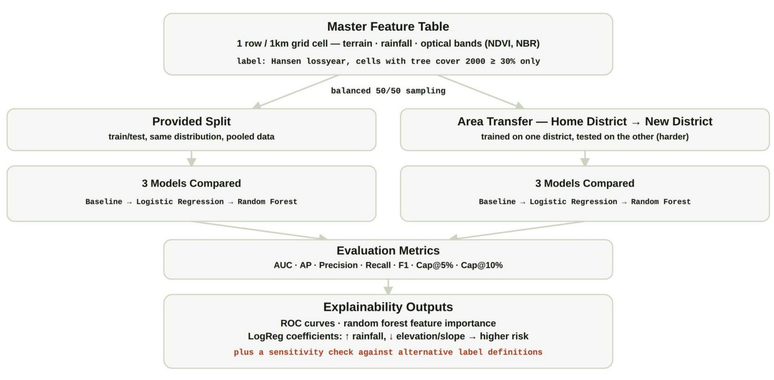
- Across Vietnam, forest rangers are tasked with patrolling vast territory to catch threats before they can spread.
- But across much of the country, forest-protection resources are stretched too thin for the area under watch — according to a 2019 World Bank assessment and field reports from provinces such as Phu Yen (Ngo Nhat, 2024)
- Patrol teams often can't respond in time to stop forest loss from illegal logging or natural disasters, per the World Bank (2019) and recent forest-loss data from Global Nature Watch (2026)



Solution Architecture



VIGIL System Pipeline



VIGIL Training pipeline

Improvement to the State-of-the-Art

- VIGIL is an interpretable AI toward a nationwide early-warning system
- Fuses 4 public sources (Hansen GFC, SRTM, CHIRPS, Sentinel-2) into a 1km grid
- Ranks the riskiest cells before loss occurs and best model reaches AUC 0.861 in real-world deployment
- Cross-district testing shows the pipeline still provides a meaningful signal beyond one area

Data Sources for VIGIL
Hansen GFC, SRTM, CHIRPS, Sentinel-2

Risk Prediction with VIGIL
Baseline, logistic regression, and random forest

Live Deployment on Public Website
VIGIL is live at www.vigil.green

Operational Integration with Local Rangers
Sending warnings to local ranger patrols

System pipeline:

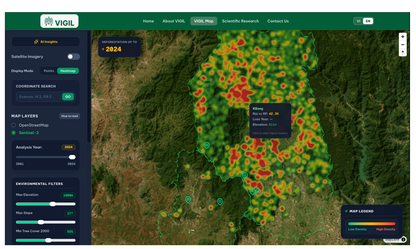
- Fuses public satellite datasets into a 1km grid
- Ranks cells into a live risk map on a province-agnostic public website
- Rangers and residents submit geolocated field reports there
- Feeding verified reports back into retraining is the next milestone

Model pipeline:

- 3 models compared—baseline, logistic regression, random forest
- Tested on a standard split and a harder cross-district transfer (train on one district, test on one never seen)
- Hardest real-world test of geographic generalization

Scaling:

- New region = only a new boundary + grid, no architecture redesign



Results

- VIGIL's first validation is in Gia Lai, where rangers cover over 1,500 hectares each
- Random Forest model reaches AUC 0.861 on in-distribution evaluation
- Area transfer test: performance drops as expected, but Logistic Regression still reaches AUC 0.589—confirms a transferable deforestation signal

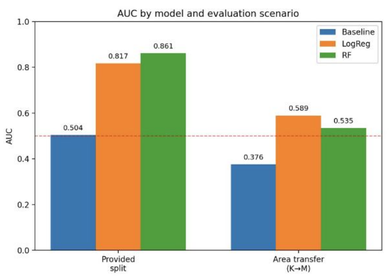


Fig. 5. AUC by model and evaluation scenario (provided split vs. area transfer, K'Bang → Mang Yang).

Model	Provided AUC	Provided Cap@5%	Transfer AUC	Transfer Cap@5%
Baseline	0.504	0.077	0.376	0.028
LogReg	0.817	0.089	0.589	0.062
RF	0.861	0.087	0.535	0.059

TABLE III. SUMMARY — AUC AND CAP@5% ACROSS SCENARIOS

GLOSSARY

Area Transfer: A harder evaluation setup where the model is trained on one district (K'Bang) and tested on a district it has never seen (Mang Yang). It measures whether VIGIL's predictions hold up beyond the area they were trained on, rather than memorizing local patterns.

AUC (Area Under the Curve): A single score summarizing how well a model ranks truly at-risk grid cells above safe ones, ranging from 0.5 (random guessing) to 1.0 (perfect ranking). VIGIL's best model reaches 0.861 under standard evaluation.

Baseline Model: A simple statistical threshold model used as a reference point. Any model VIGIL deploys must clearly outperform this baseline to justify its added complexity.

Cap@5%: The share of all truly deforested grid cells that fall within the top 5% of cells the model ranks as highest-risk. It reflects how useful the risk map is when only a limited number of patrols can be sent out.

Feature Engineering: The process of converting raw satellite and terrain data into structured variables — slope, rainfall, vegetation indices — that a model can learn from.

Feature Importance: An explainability output from the Random Forest model showing which input variables most influence its predictions, helping rangers understand why a cell was flagged.

Grid Cell: VIGIL's smallest unit of prediction — a 1km × 1km tile of land. Each cell carries its own row of features and a risk score.

Hansen GFC (Global Forest Change): A public global dataset tracking year-by-year forest loss from satellite imagery. VIGIL uses it both as a historical feature and as the ground-truth label for training.

Logistic Regression: An interpretable linear model whose coefficients directly show how each feature raises or lowers predicted risk. It generalizes better than Random Forest when tested on an unseen district.

Province-Agnostic Architecture: A system design that works for any province without being rebuilt — adding a new region only requires loading its administrative boundary and 1km grid, not redesigning the pipeline or platform.

Random Forest: A nonlinear ensemble model combining many decision trees. It achieves VIGIL's highest accuracy on standard evaluation but loses more ground than Logistic Regression under area transfer.

ROC Curve: A plot of a model's true-positive rate against its false-positive rate across all risk thresholds. The area beneath this curve is the AUC score.

Sensitivity Check: A robustness test that re-runs the analysis using an alternative definition of "forest loss" to confirm the main conclusions don't depend on one labeling choice.